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Interview: Why is Redis so fast even though it is single-threaded? Please state your insights on Redis' thread model

Dylan Smith · Follow

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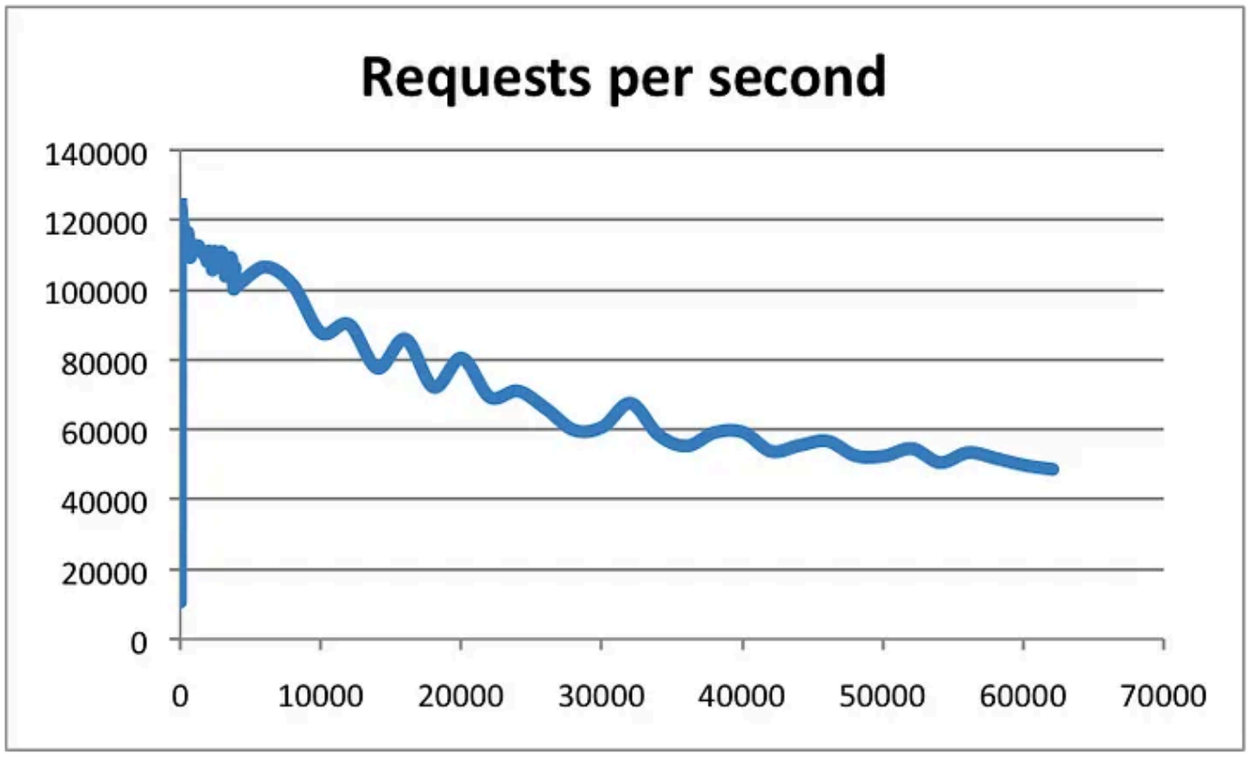
How fast is Redis?

Redis is a high-performance key-value database that runs in memory. The test report provided by the official is that a single machine can support about 100,000 QPS (Queries Per Second). 😱



Why is Redis so fast ❓

But we have also heard that Redis is single-threaded. Why is Redis, which uses a single thread, so fast? In this article, we will analyze the reasons for this.

In fact, strictly speaking, the Redis Server is multi-threaded, but its entire request processing flow is handled by a single thread. We must clearly understand this point and not simply think that the entire Redis Server is single-threaded!

When we say that Redis is fast with a single thread, we mean that its request processing process is extremely fast!

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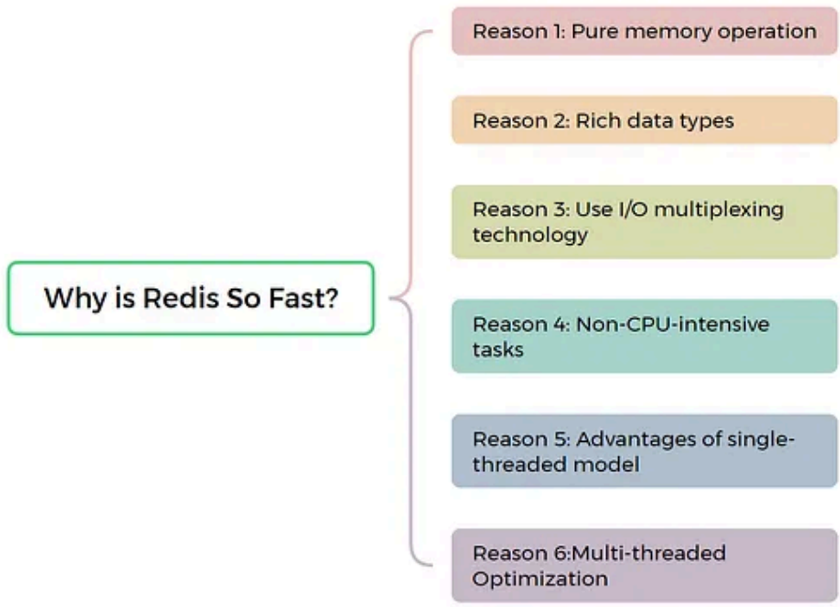
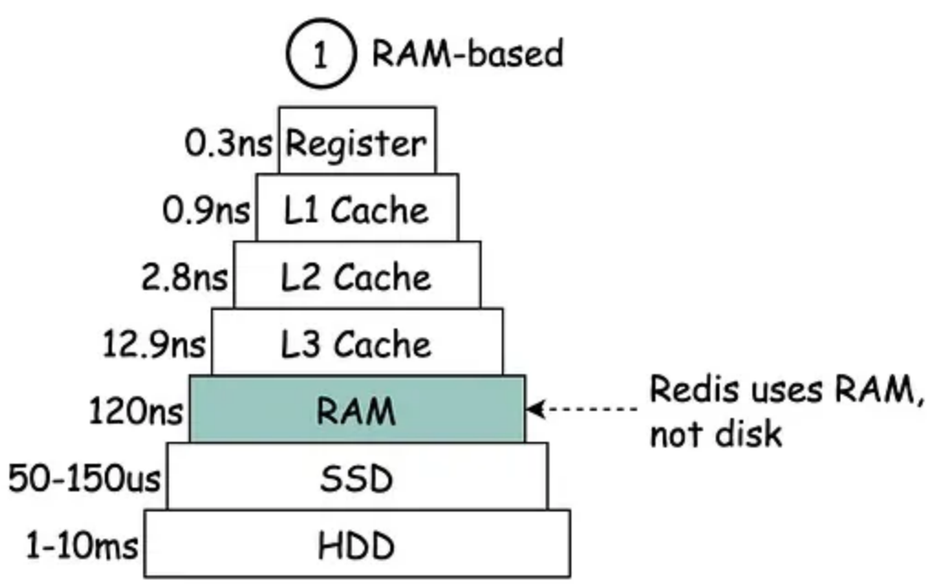
Now let's analyze why using a single thread for request processing can still achieve such high performance. Its excellent performance mainly depends on the following aspects:



Reason 1: Pure memory operation

Redis is an in-memory database. Its data is all stored in memory, which means that our reading and writing of data are all completed in memory. This speed is very fast.



Not only that, Redis is also a key-value in-memory database. It internally constructs a hash table. When accessing according to a specified KEY, the corresponding data can be found with only O(1) time complexity.

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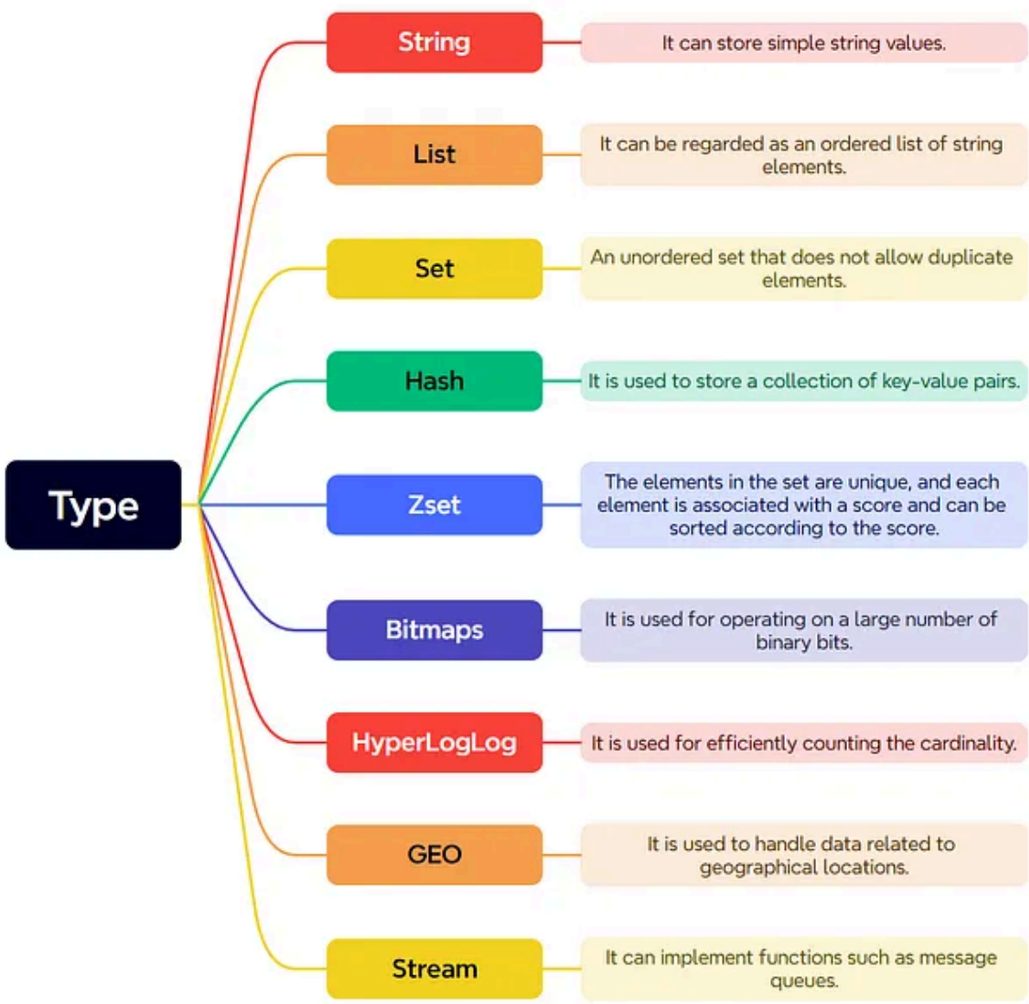
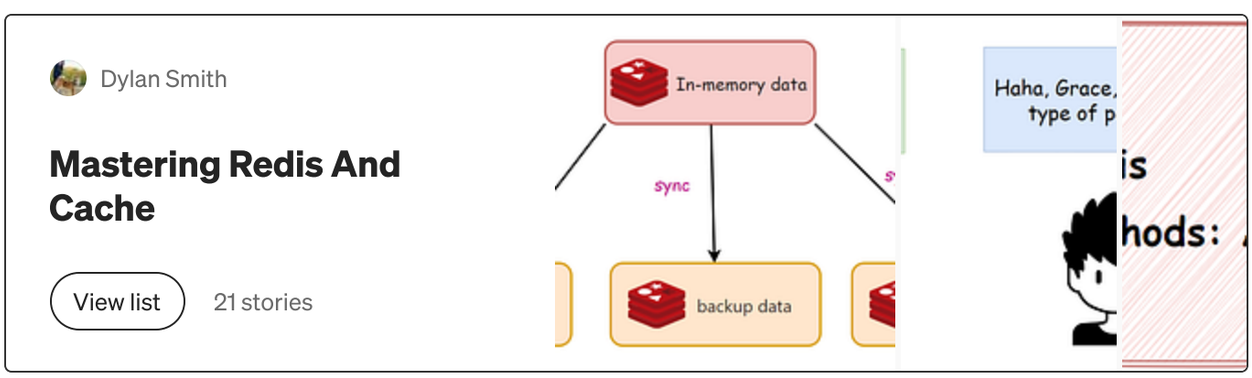
Reason 2: Rich data types

Redis provides rich data types. You can use different data types for efficient operations according to different scenarios. These operations are all performed in memory and do not consume a large amount of CPU resources, so the speed is extremely fast.



Regarding the detailed introduction of each data type, you can refer to my column:



And, if you need resources, you can also refer this article

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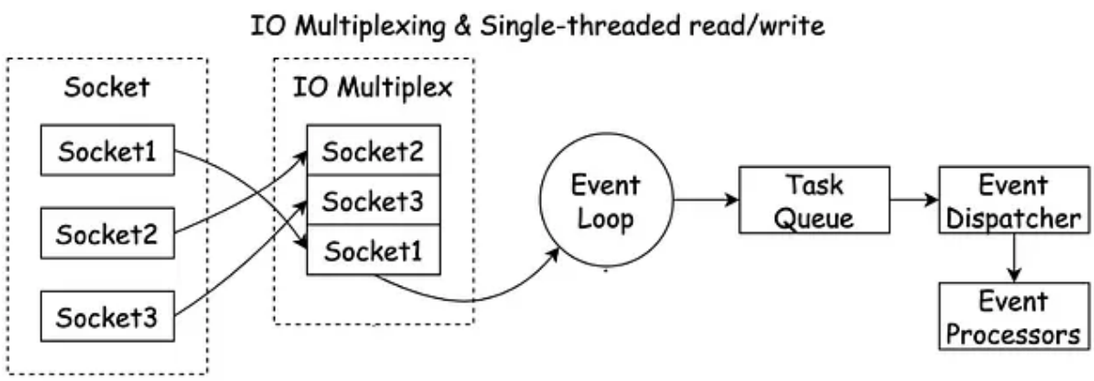
Reason 3: Use I/O multiplexing technology

Redis uses a single thread, so how does it handle multiple client connection requests?

Redis uses I/O multiplexing technology and non-blocking I/O. This technology is provided by the operating system implementation. Redis can conveniently use the operating system's API.

> The so-called I/O multiplexing (I/O Multiplexing) is actually a synchronous I/O model used to simultaneously monitor multiple file descriptors (File Descriptor) to determine which one or which descriptors can perform I/O operations (such as reading, writing, etc.).
>
> Its core idea is that through a system call (such as select, poll, epoll, etc.), the program can simultaneously block and wait for the ready state of multiple I/O operations. When one or more descriptors are ready, the system call returns and tells the program which descriptors are ready for I/O operations. Then the program can perform corresponding I/O processing for these ready descriptors.
>
> Compared with the traditional blocking I/O model, I/O multiplexing can handle multiple I/O operations in a single thread, avoiding creating a thread for each I/O operation, thereby improving the utilization rate and concurrency of system resources.

Because of this, Redis can listen to requests from multiple Sockets in a single thread. When any Socket is readable/writable, Redis reads the client request, operates the corresponding data in memory, and then writes it back to the Socket.



The whole process is extremely efficient. Redis utilizes the event-driven model of I/O multiplexing technology to ensure that it only reacts to active Sockets when monitoring multiple Socket connections.

Reason 4: Non-CPU-intensive tasks

The disadvantage of using a single thread is obvious. It cannot utilize multi-core CPUs. The author of Redis mentioned that since most operations of Redis are not CPU-intensive tasks (referring to those tasks that mainly rely on CPU computing power to complete), and the bottleneck of Redis lies in memory and network bandwidth.

Under high-concurrency requests, Redis requires more memory and higher network bandwidth. Otherwise, the bottleneck is likely to occur in situations where there is insufficient memory or network latency waiting.

Of course, if you think the performance of a single Redis instance is insufficient to support business, the author of Redis recommends deploying multiple Redis nodes and forming a cluster to utilize the capabilities of multi-core CPUs instead of using multiple threads to process on a single instance.

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Reason 5: Advantages of single-threaded model

Based on the above characteristics, Redis using a single thread is already sufficient to achieve very high performance. However, the single-threaded model itself has the following advantages:

- There is no performance loss due to multi-thread context switching.
- There is no performance loss due to the need for locking when accessing shared resources.
- It is very friendly for development and debugging and has high maintainability.

Based on these aspects, Redis finally adopts a single-threaded model to complete the entire request processing work.

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Multi-threaded Optimization

As already specifically stated at the beginning of the article, **Redis Server itself is multi-threaded.**

Apart from the request processing flow being handled by a single thread, there are other worker threads inside Redis that execute in the background. They are responsible for asynchronously executing certain relatively time-consuming tasks. For example, AOF flushing every second and AOF file rewriting are completed in another thread.

After Redis 4.0, Redis introduced the `lazyfree` mechanism and provided commands such as `unlink`, `flushall async`, `flushdb async`, and mechanisms such as `lazyfree-lazy-eviction` and `lazyfree-lazy-expire` to release memory asynchronously. This is mainly to solve the performance problem of the entire Redis being blocked when releasing large memory data.

When deleting large keys, releasing memory is often time-consuming. Therefore, Redis provides an asynchronous way to release memory. Putting

these time-consuming operations in another thread for asynchronous processing does not affect the execution of the main thread and improves performance.

In Redis 6.0, Redis introduced multi-threading again to complete the protocol parsing of request data to further enhance performance. It mainly addresses the pressure brought by single-threaded protocol parsing of request data in high-concurrency scenarios.

After the protocol parsing of request data is completed by multiple threads, the subsequent request processing stage is still handled in a single-threaded queue.

It can be seen that Redis does not conservatively think that a single thread is so good, nor does it introduce multi-threading just for the sake of using multi-threading. The author of Redis is very clear about the usage scenarios of single-threaded and multi-threaded models and makes targeted optimizations. This is very worthy of our learning.

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Disadvantages

The above introduction shows that a single thread can achieve such high performance, but it does not mean that it has no disadvantages.

The biggest disadvantage of single-threaded processing is that if a previous request takes a long time to operate, then the entire Redis will be blocked and other requests cannot enter until this time-consuming operation is completed and returned. Only then can other requests be processed.

When we encounter problems of Redis slowing down or being blocked for a long time, more than 90% are caused by the blocking of Redis' single thread.

Therefore, when using Redis, we must avoid very time-consuming operations.

For example, obtaining data in a way with too high time complexity, obtaining too much data at one time, and a large number of keys expiring simultaneously leading to increased pressure on Redis to eliminate keys. These scenarios will block the entire processing thread until they are completed, which will inevitably affect business access.

For the content introducing specific scenarios that can cause Redis to be blocked and providing methods to avoid problems and optimization schemes, you can refer to my article:

Interview: 11 Tips for Redis to Achieve High Performance

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Finally, if the article was helpful, please clap 👏 and follow, thank you! ┗(*°▽°*)┛

I'm Dylan, looking forward to progressing with you. ❤️

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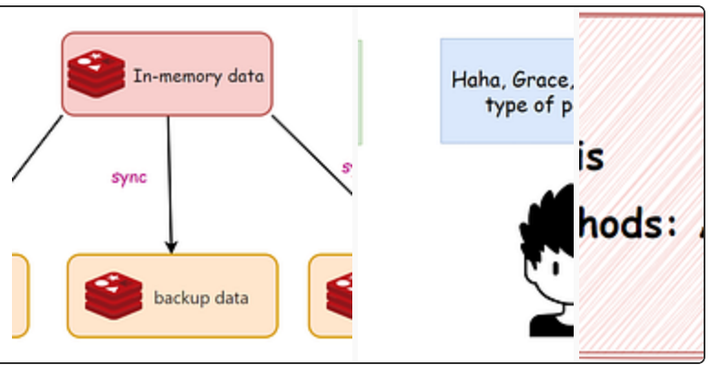

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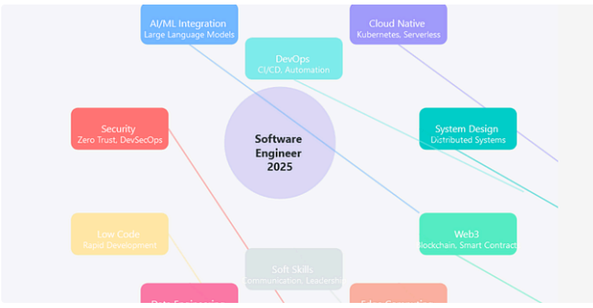

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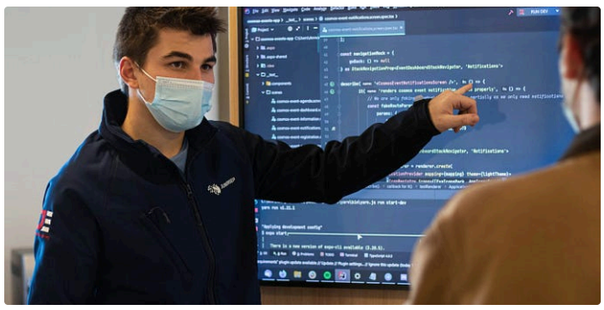

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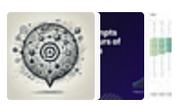

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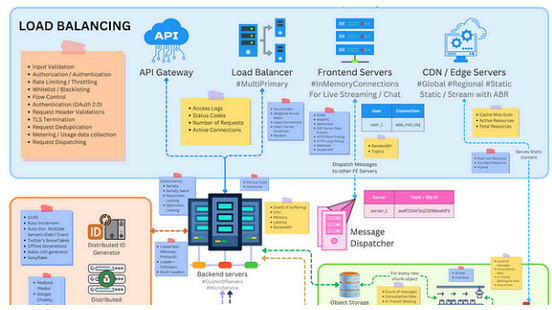

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